

YEAR 9 SCIENCE
CHEMICAL SCIENCES
WORKSHEET - ACIDS, BASES AND SALTS

1. Determine whether the following substances are *ACID (A)*, *BASE (B)* or *SALT (S)*. Write the appropriate letter next to the substance.

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|------------------------------|-----------------------------------|--|--|
| (a) S NaCl | (b) B MgO | (c) B zinc hydroxide | (d) A HNO ₃ |
| (e) S copper sulphate | (f) B copper oxide | (g) S Al ₂ (SO ₄) ₃ | (h) A H ₂ SO ₄ |
| (i) B NaOH | (j) A HCl | (k) S FeCl ₃ | (l) S Ca(NO ₃) ₂ |
| (m) B sodium oxide | (n) A CH ₃ COOH | (o) S potassium acetate | |

2. Complete the following general word equations for acid reactions.

ACID + METAL → SALT + HYDROGEN (H₂)

ACID + BASE → SALT + WATER (H₂O)

ACID + CARBONATE → SALT + WATER (H₂O)
+ CARBON DIOXIDE (CO₂)

3. For the following acid reactions, *complete the word equations*. Use the general equations in Q2 above to help you.

- (a) magnesium oxide + sulphuric acid → **magnesium sulphate + H₂O**
base
- (b) aluminium + hydrochloric acid → **aluminium chloride + H₂**
metal
- (c) sodium carbonate + sulphuric acid → **sodium sulphate + H₂O + CO₂**
- (d) potassium hydroxide + nitric acid → **potassium nitrate + H₂O**
base
- (e) calcium + acetic acid → **calcium acetate + H₂**
metal
- (f) copper oxide + acetic acid → **copper acetate + H₂O**
base
- (g) iron carbonate + hydrochloric acid → **iron chloride + H₂O + CO₂**
- (h) zinc + sulphuric acid → **zinc sulphate + H₂**
metal
- (i) calcium carbonate + nitric acid → **calcium nitrate + H₂O + CO₂**